

Adnan Mashrur Sadad

+60 11-3968 7435 | adnanmashrursadad@gmail.com | LinkedIn | GitHub | Portfolio

Professional Summary

Software engineer specializing in applied AI/ML and GenAI systems — from real-time streaming pipelines and drift-aware retraining to production model serving (Kafka, Kubernetes, FastAPI, Docker, CI/CD). Builds with the rigor production systems demand: honest time-ordered validation, automated drift detection, and retraining pipelines that recovered model performance by 98% within a single cycle. B.Sc. Computer Science (Software Engineering), Universiti Teknologi Malaysia.

Technical Skills

- **GenAI / LLM Systems:** RAG, Agentic Workflows, Text2SQL, Prompt Engineering, Structured Output Validation, LLM Evaluation & Repair Loops, Groq API, Whisper
- **Machine Learning & Data:** XGBoost, Scikit-learn, TensorFlow, Pandas, NumPy, Computer Vision (CNN/ANN), OCR, Predictive Analytics, Monte Carlo Simulation, Hugging Face
- **Programming Languages:** Python, Java, C++, JavaScript, TypeScript, SQL, Dart
- **Frameworks & Tools:** FastAPI, React, SQLAlchemy, Streamlit, Recharts, Spring, Flutter, Git
- **Deployment & Architecture:** Docker, REST APIs, Microservices, CI/CD (GitHub Actions), OpenAPI
- **Core CS:** Data Structures & Algorithms, OOP, Operating Systems, Databases, Software Engineering

Projects

InterviewPilot – AI-Powered Mock Interview Platform 2026 – GitHub

Python, FastAPI, SQLAlchemy, React + TypeScript, Groq API (Whisper-large-v3, Llama 3.3 70B), Docker, GitHub Actions

- Architected a full-stack mock interview platform (FastAPI/SQLAlchemy backend, React + TypeScript frontend) using Whisper-large-v3 for speech transcription and Llama 3.3 70B for automated interview evaluation
- Built a rubric-based evaluator that returns schema-validated JSON (technical accuracy, clarity, depth) with a self-correcting repair loop that re-prompts the model on its own invalid output
- Designed an agentic follow-up question generator that reasons over candidate answers to decide when to probe deeper, mirroring real interviewer behavior
- Enforced engineering rigor with a pytest suite (17 tests, 87% backend coverage, fully mocked Groq calls), Dockerized deployment, and GitHub Actions CI on every push

Driftline – Real-Time Fraud Detection & Drift-Aware ML Platform 2026 – GitHub

Python, XGBoost, PyTorch Geometric, Apache Kafka (Redpanda), PyFlink, Feast, FastAPI, ONNX, Docker, Kubernetes, MLflow, GitHub Actions

- Built a streaming fraud-detection pipeline replaying 590K+ transactions via Apache Kafka (Redpanda) and PyFlink, computing real-time entity-velocity features into a Feast feature store at 150+ events/sec
- Trained an XGBoost classifier on a strict time-ordered holdout (118K transactions, 0.48 PR-AUC) and engineered a PyTorch Geometric GraphSAGE model over a 605K-node identity graph with a unit-tested leakage boundary
- Designed an automated drift-detection and retraining system (weekly PSI/KS monitoring across 369 features) that recovered model PR-AUC by 98% within one retrain cycle via shadow-evaluation and an enforced promotion gate
- Deployed the scorer as an ONNX-optimized FastAPI service on Kubernetes with Prometheus/Grafana observability, validated by a 5-stage GitHub Actions CI/CD pipeline and load-tested to a live autoscaling event

AI Chatbot Engineer Assessment — Mindhive Asia 2025 – GitHub

FastAPI, RAG, Text2SQL

- Designed a multi-turn conversational AI with stateful memory and intent-based agentic planning, sustaining coherent context across 3-5 sequential user turns
- Built FastAPI microservices including a RAG-powered product knowledge endpoint and a Text2SQL outlet-query endpoint, validated against 200+ product documents with SQL-injection protection
- Evaluated retrieval quality with a hand-labeled 20-query harness against the live product/outlet corpus: 85% Hit Rate@3 overall (100% on outlet lookups), MRR 0.86
- Delivered a complete repository with OpenAPI specification, test suite, architecture diagrams, and a hosted live demo

WC26 Predictor – World Cup 2026 Forecasting Dashboard 2026 – GitHub

Python, XGBoost, FastAPI, React, TypeScript, Recharts

- Built an end-to-end football analytics pipeline using historical match results, FIFA rankings, rolling form, squad proxies, and head-to-head features to model match outcomes and expected goals
- Benchmarked outcome models on held-out matches: XGBoost reached 59.4% accuracy (vs. 47.8% majority-class baseline) with a calibrated 0.522 Brier score, and a Poisson goals regressor achieved 1.02 MAE
- Ran Monte Carlo simulations over the calibrated probabilities to project group qualification, knockout paths, and championship odds, surfaced in an interactive dashboard with tournament-path curves and a predicted-vs-actual audit view

Work Experience

Joget Inc. Oct 2024 – Mar 2025

Software Developer Intern (Hybrid)

- Built and deployed enterprise workflow automations and UI components on the Joget DX low-code platform, digitizing core business processes for clients across multiple departments
- Integrated REST APIs with external services, streamlining data flow and reducing manual processing effort by 30%
- Shipped production features within Agile sprints alongside senior engineers, QA, and product managers, covering the full SDLC from design through deployment
- Deployed and maintained workflow modules through CI/CD pipelines, contributing to recurring production release cycles for enterprise clients

Education

Universiti Teknologi Malaysia (UTM), MJIIT 2021 – 2026

Bachelor of Computer Science (Software Engineering) with Honours, CGPA: 3.43/4.00 — Dean's List 2022/2023, 2025/2026 Kuala Lumpur, Malaysia

- Undergraduate thesis: OCR-based personal finance tracking system for automated income-tax-ready expense classification
- Relevant coursework: Artificial Intelligence & Machine Learning, Software Engineering, Data Structures & Algorithms, Operating Systems, Computer Architecture

Maple Leaf International School

A Levels & O Levels

Dhaka, Bangladesh

Achievements & Leadership

- Gold Medal & Best Video Award, myHCI-UX Student Design Challenge (Oct 2025) — national university competition for UX design and innovation
- Secretary Treasurer, SOFEA Society (2022–2023); organized AI workshops for 200+ students